

STAT 812/420

Computational Statistics

1 3 4 8

$$= 1 \times 10^3 + 3 \times 10^2 + 4 \times 10 + 8 \times 10^0$$

1 2 4 5 6

10^4 10^3 10^2 10^1 10^0

1 1 1 0 1

$$1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^1 + 1 \cdot 2^0$$

$$0 \ 0 \ 0 = 0$$

$$0 \ 0 \ 1 = 1$$

$$0 \ 1 \ 0 = 2$$

$$0 \ 1 \ 1 = 3$$

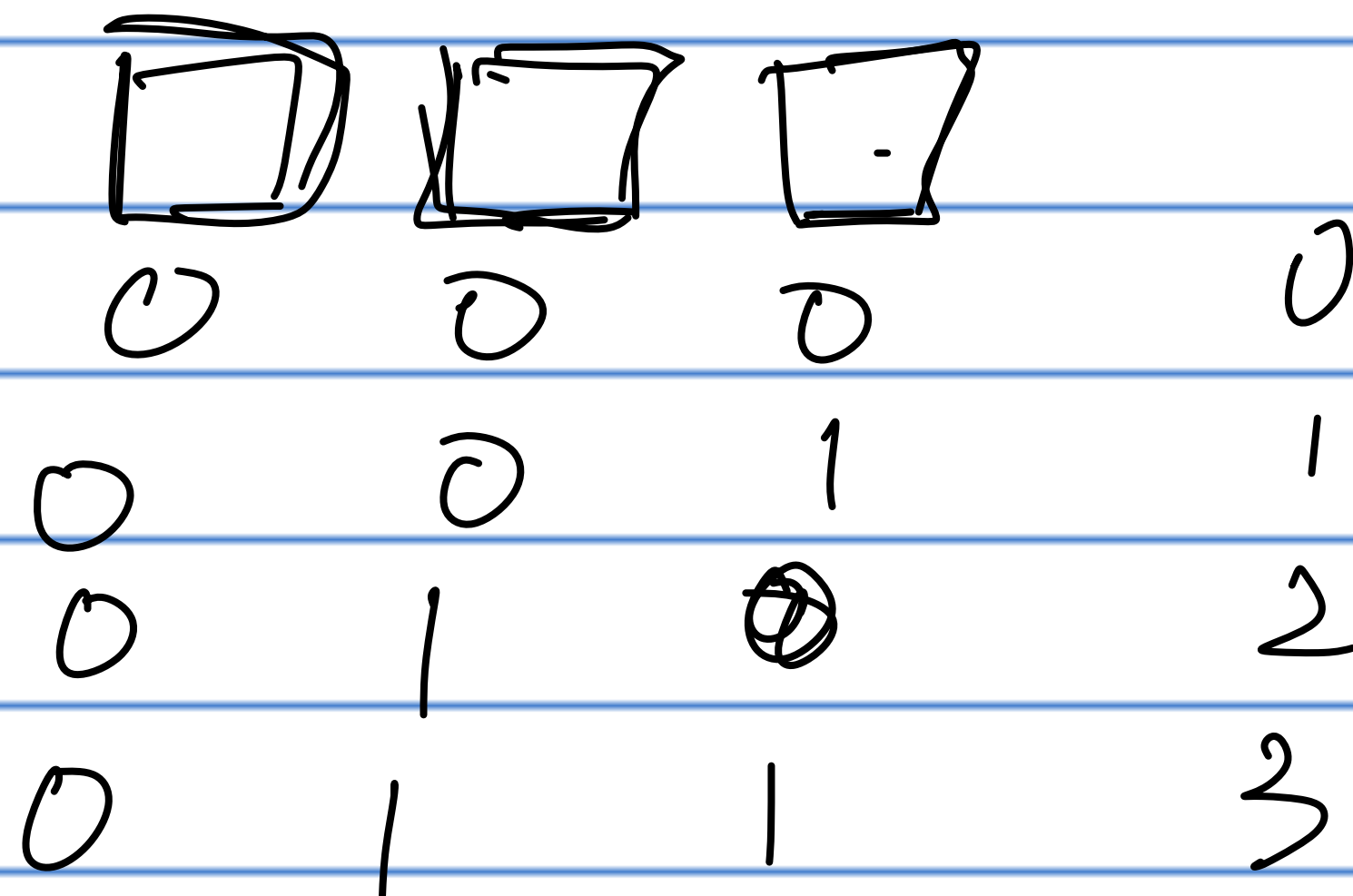
$$1 \ 0 \ 0 = 4$$

Integer

$\boxed{\emptyset/1}$ $\boxed{0/1}$ $\boxed{x/1}$

32 bits

3 bits



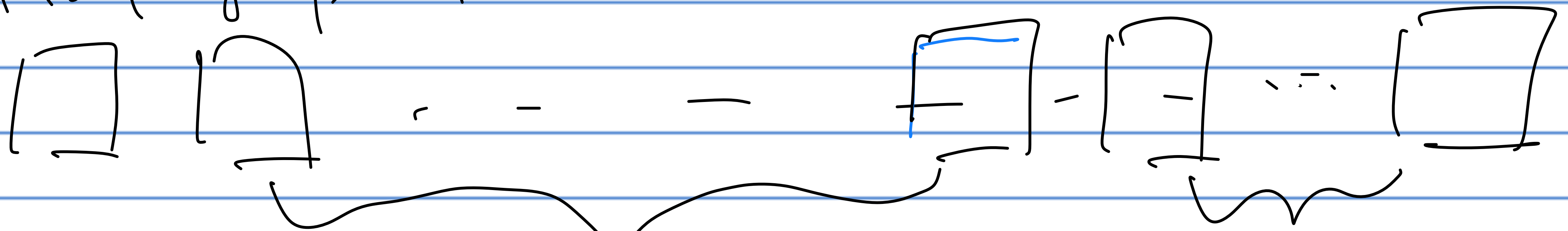
1 2 3 4
↓ ↓ ↓

$$1 \times 10^3 + 2 \times 10^2 + 3 \times 10^1 + 4 \times 10^0$$

$$\sum_{i=0}^{p-1} 2^i = 2^p - 1$$

1

Floating point



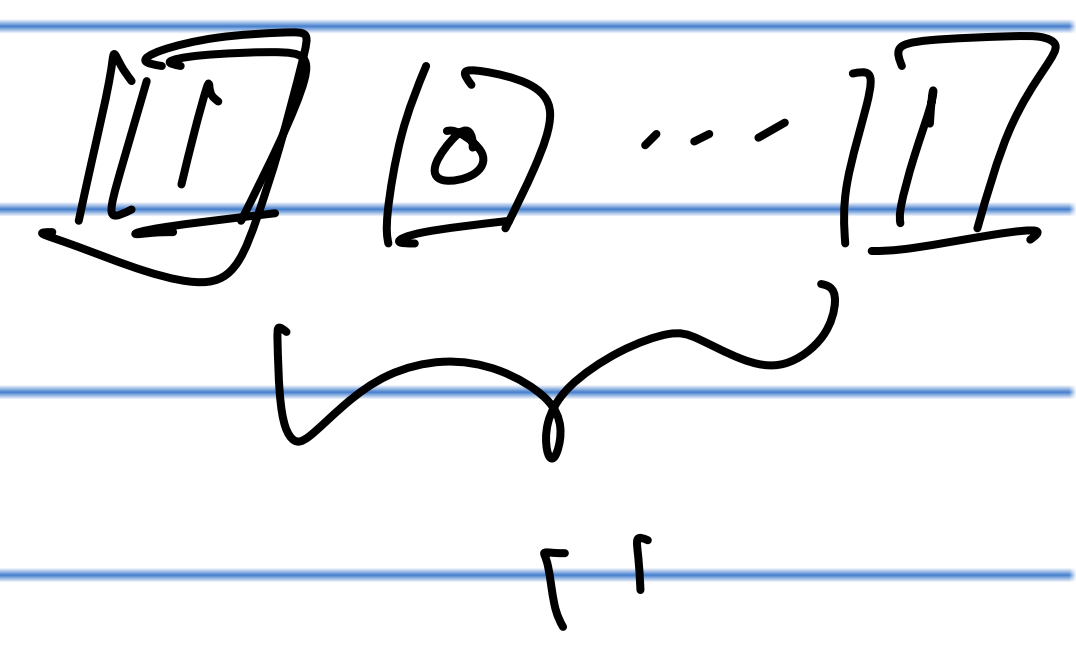
2

Sign
↓

Exp.

Mantissa

64

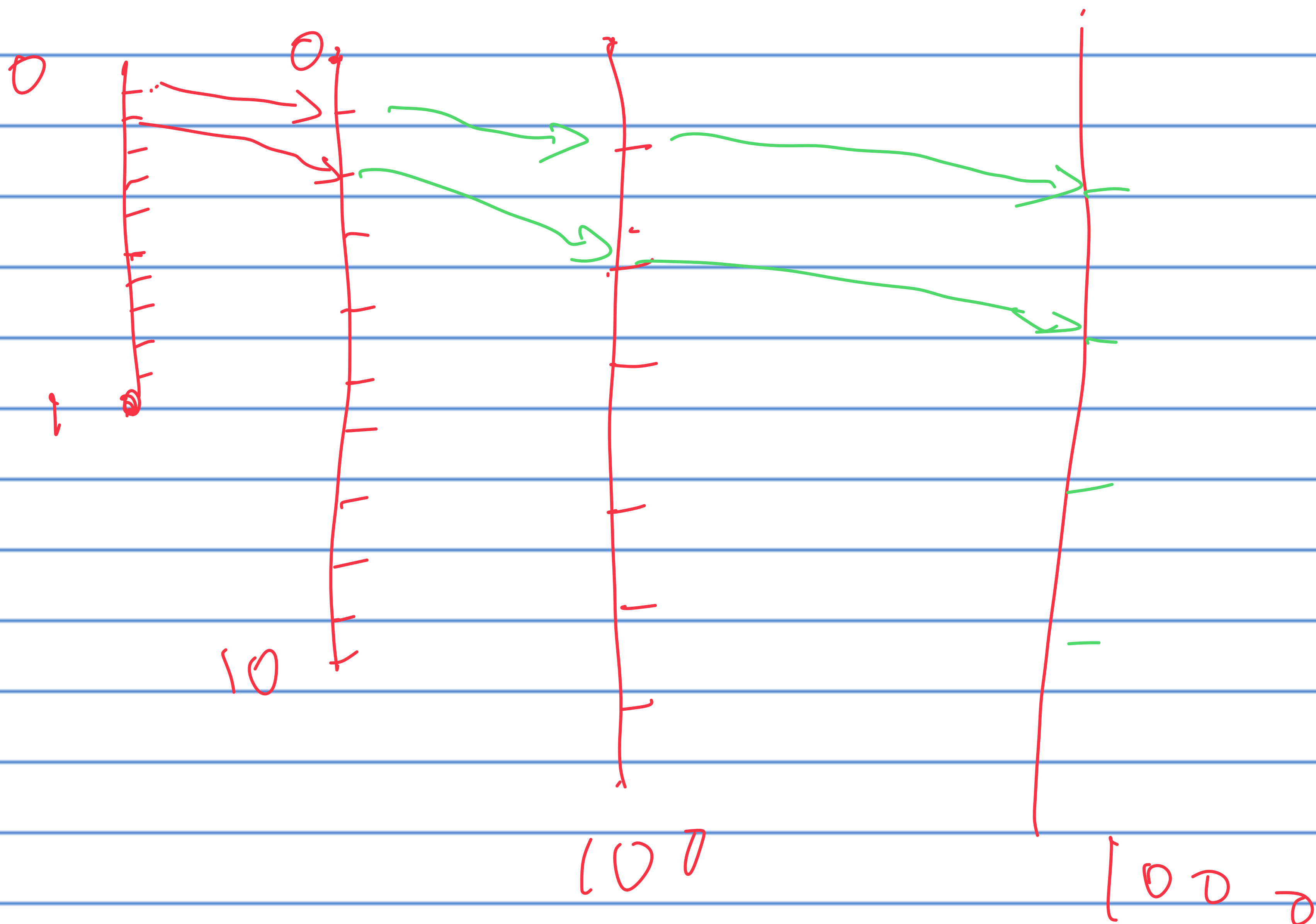


$(-1)^S$

0.010111 x 2

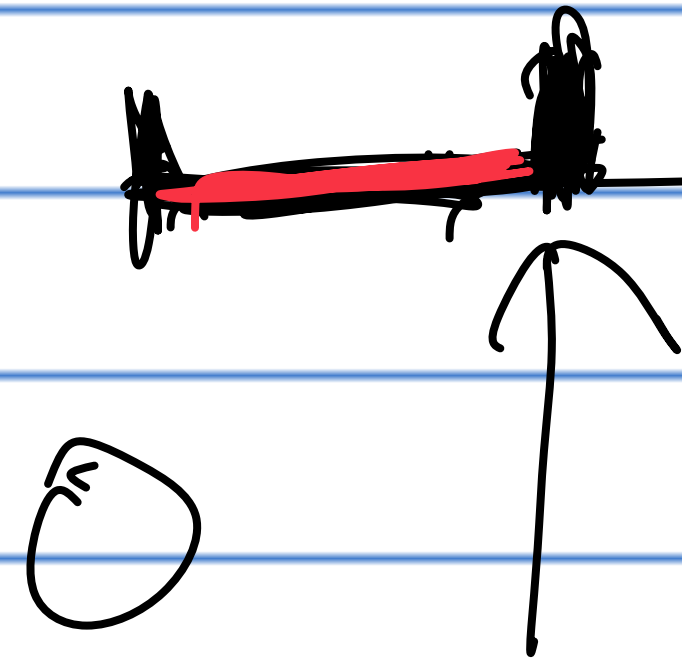
52

A Model for Understanding Floating Point Numbers

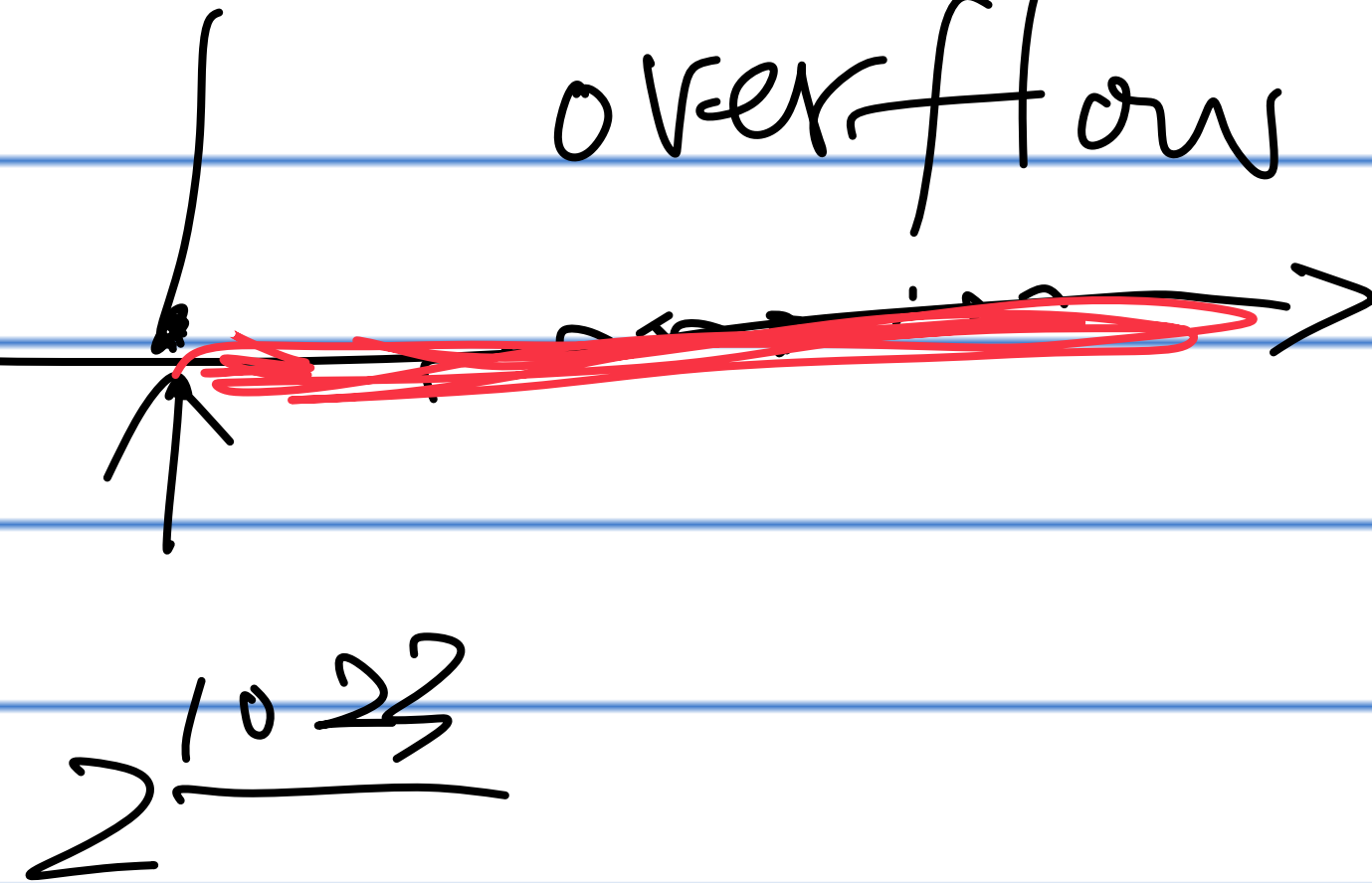


Overflow & Underflow

underflow



overflow

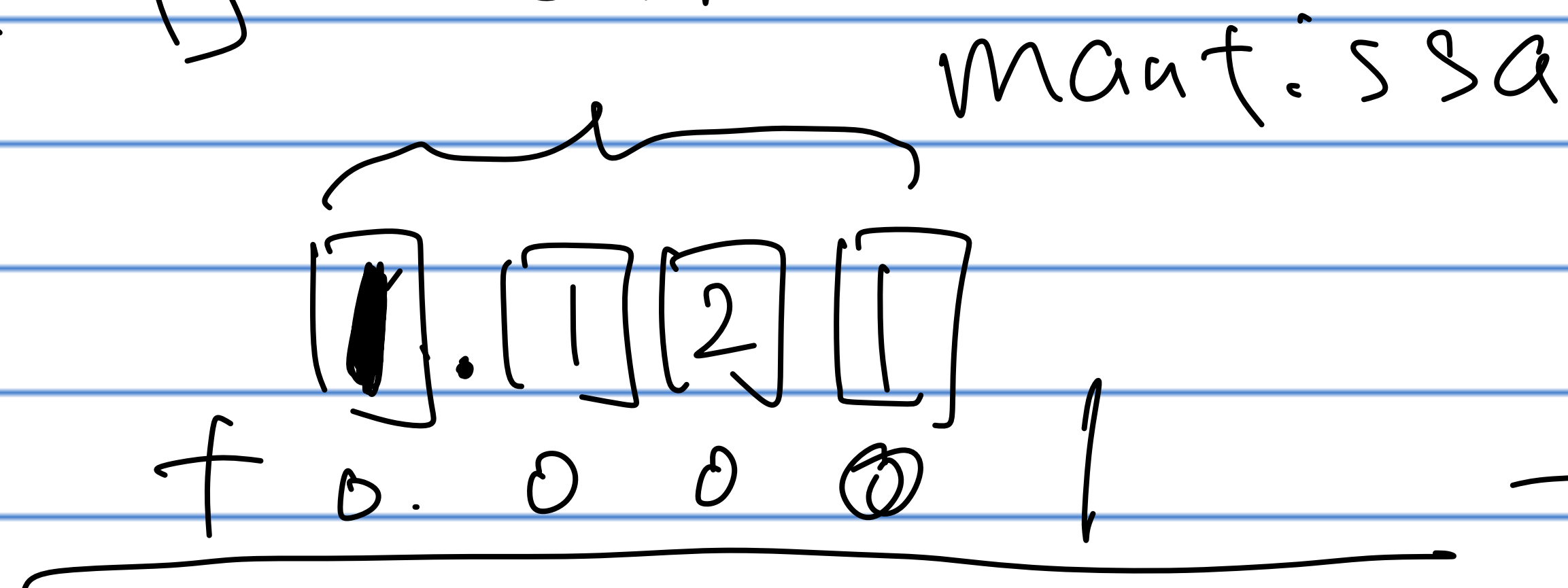


$$2^{-1074}$$
$$\approx 10^{-300}$$

$$2^{1023}$$
$$\approx 10^{300}$$

$$2^{-2000} \rightarrow 0$$

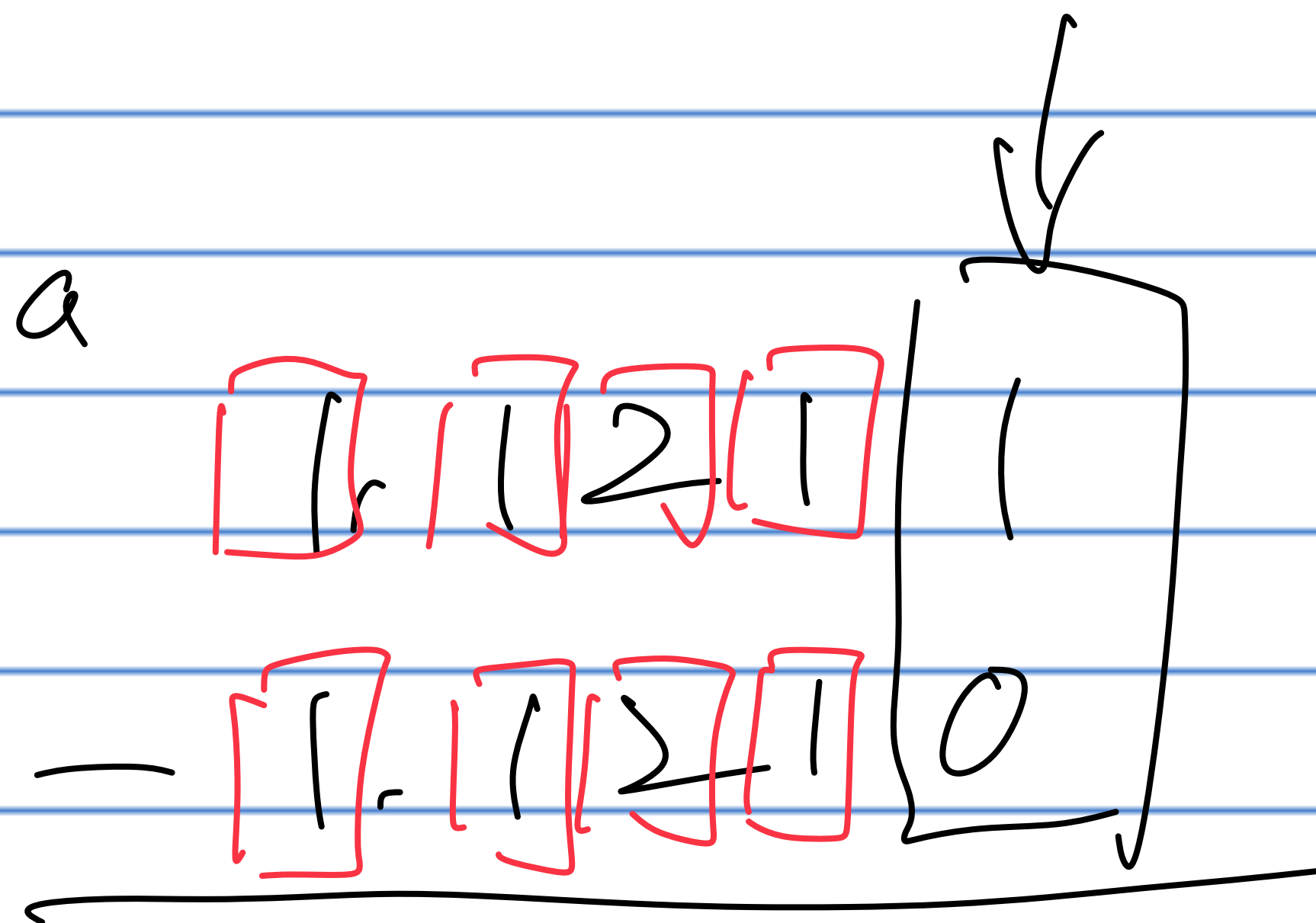
Round off error



1.121 ~~1~~



1.121



0.0000

$\neq$

0.0001 $\leftarrow$

Example

$$1 + 2^{-40}$$

$$\begin{array}{r} 1.000\dots 0 \\ \hline + 0.000\dots 01x \end{array}$$

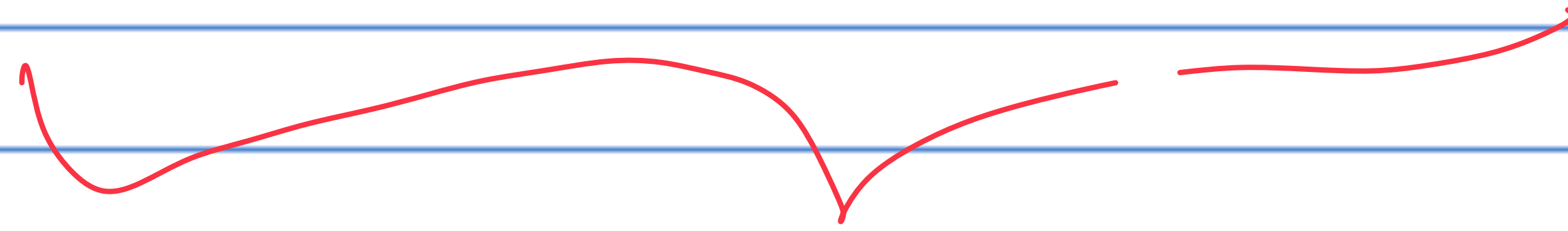
$$1 \boxed{+} 2^{-53} \boxed{+} \dots \boxed{+} 2^{-53}$$

$$1 + 2^{-52} \times 2^{12} = 1 + 2^{-40}$$

$\boxed{2} \cdot \boxed{0}$

$\dots$

$\boxed{0}$



$\rightarrow 1$

$+ 0 \cdot 0$

$\dots$

$\boxed{0}$

$1 = 2^{-52}$



$1 \cdot 0$

$\dots$

0

$\times$

$$2^{-54} + 2^{-54} = 2 \cdot 2^{-54}$$

4 bits in base 10

$$\underline{1.000} + 0.0001 + 0.0001$$

$$= 1.000 \times 10^0 + 1.000 \cdot 10^{-4} + 1.000 \times 10^{-4}$$

$$= 1.000 + 2.000 \cdot 10^{-4}$$

Example

$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^k}{k!}$$

$x < 0$

n

+ - + - + -

$x > 0$

$$e^{-x}$$

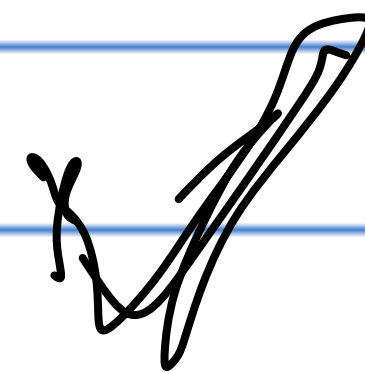
=

$$\frac{1}{e^x}$$

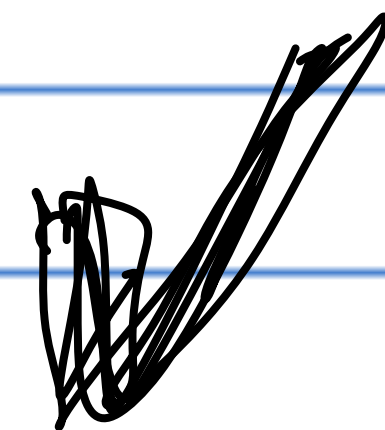
↑

Example

$$\sum_{i=1}^n (x_i - \bar{x})^2$$



$$= \sum_{i=1}^n x_i^2 - n \cdot \bar{x}^2$$



$$= \sum_{i=1}^n x_i^2 - \frac{(\sum x_i)^2}{n}$$

x_i	x_i^2
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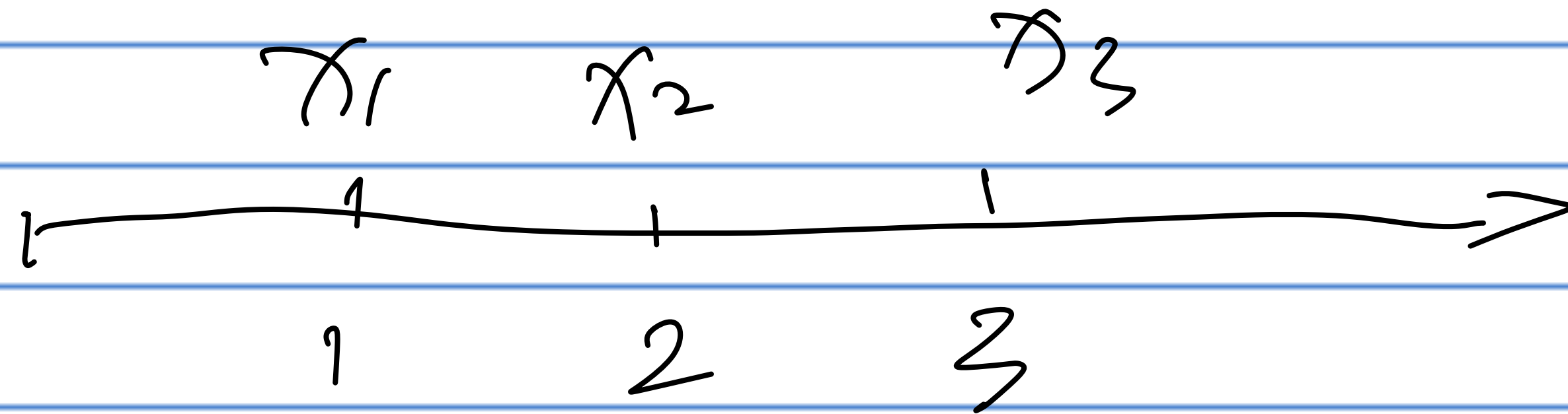
x	x
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x	x
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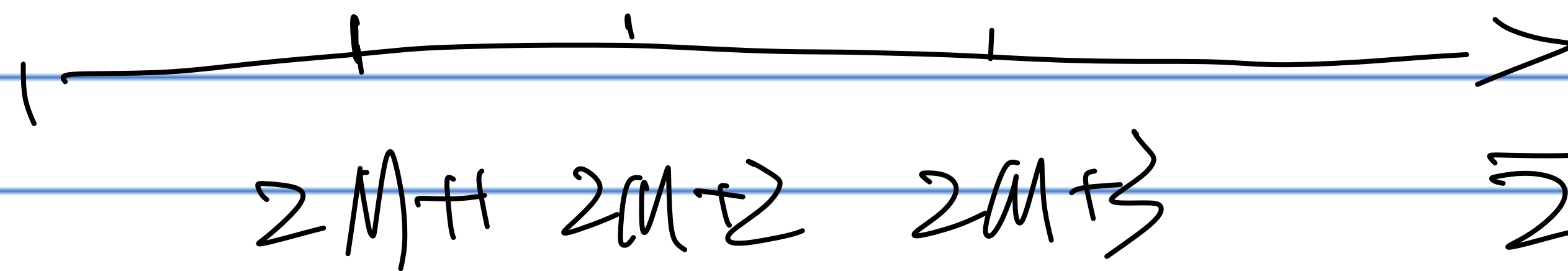
x	x
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x	x
-----	-----

$\sum x_i$	$\sum x_i^2$
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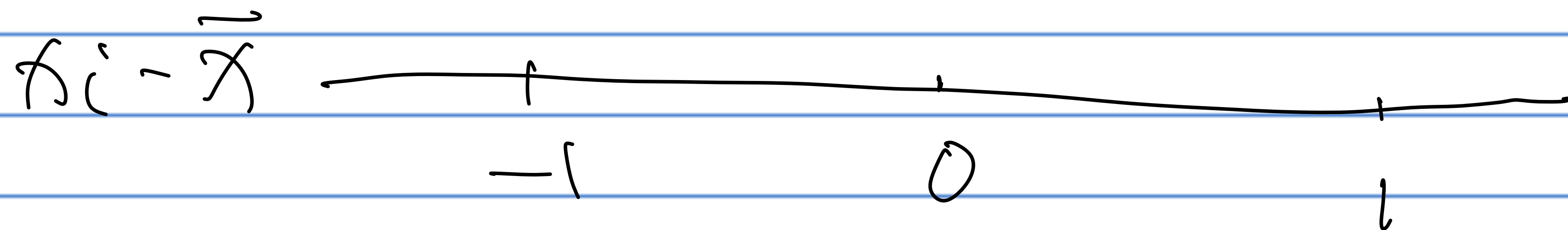
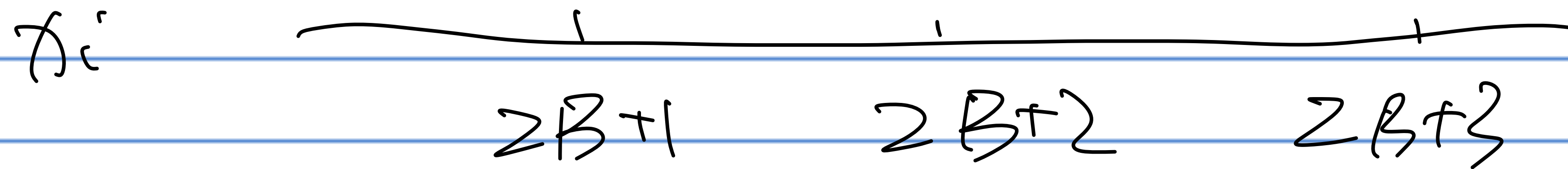
$$\sum (x_i - \bar{x})^2 = 2$$



$$\sum (x_i - \bar{x})^2 = 2$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ (2M+1)^2 & & (2M+3)^2 \end{array}$$

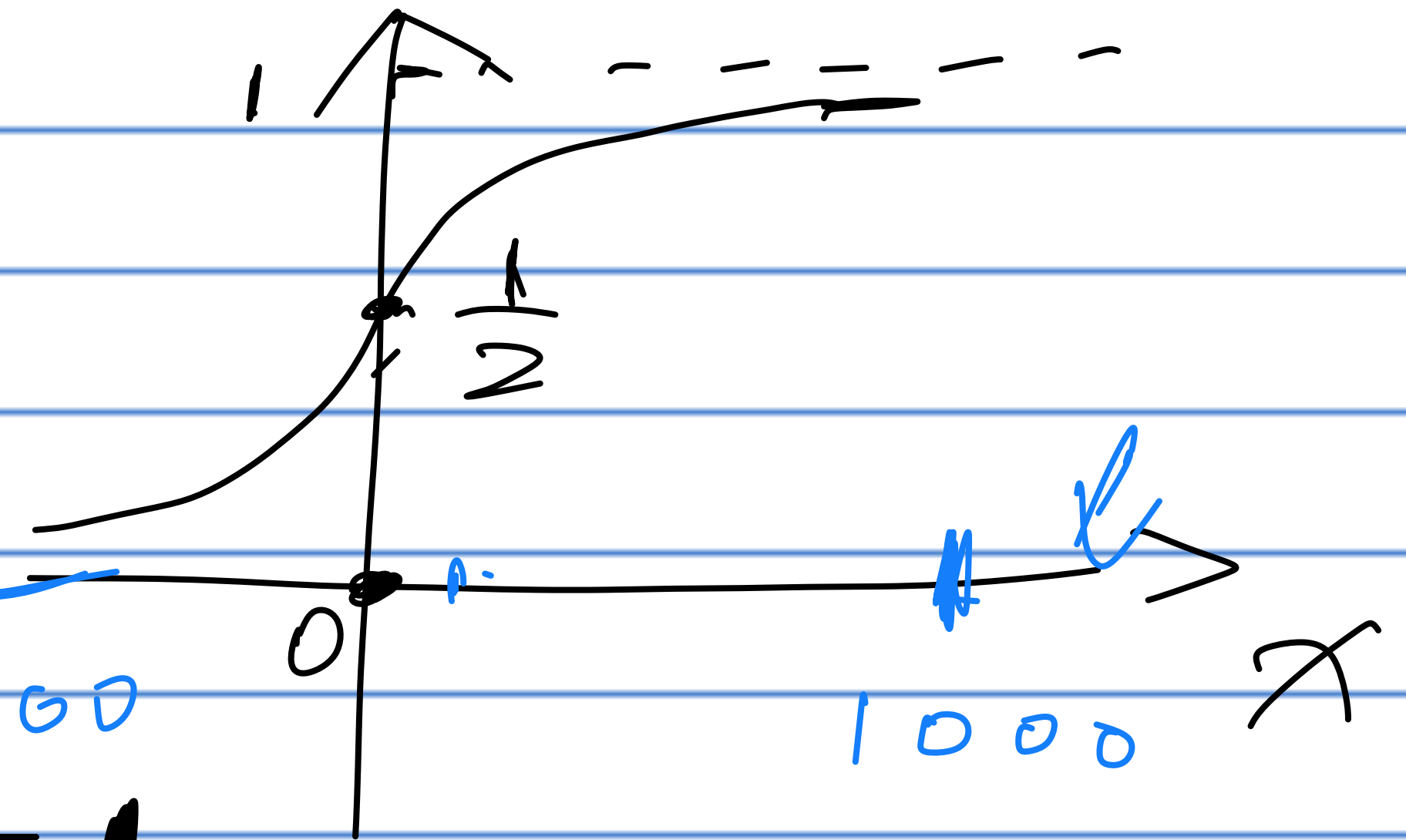
$$\left(\sum x_i \right)^2 = (6M+6)^2 \approx 12M^2 = 12 \cdot 10^{12}$$



Exaplo

$$\frac{e^x}{1+e^x} = 1$$

$$\frac{1}{1+e^{-x}}$$



$$\frac{1}{1+e^{-1000}} = 1$$

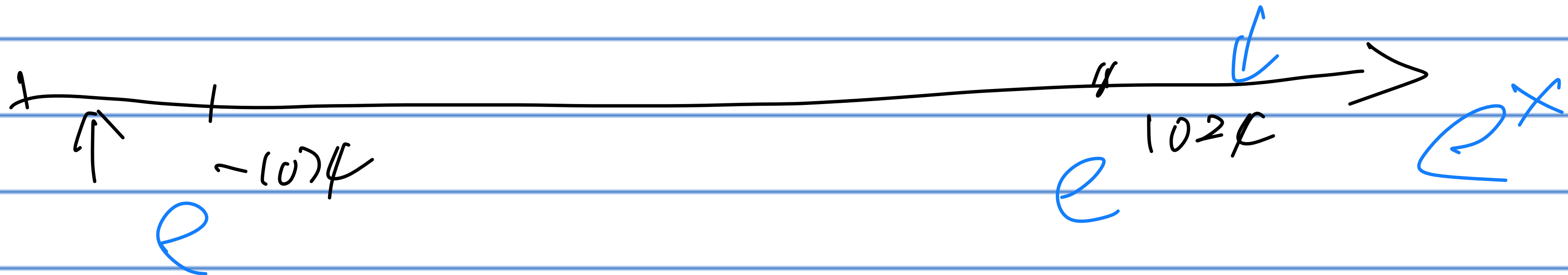
$$\frac{1}{1+\text{Inf}} = 0$$

e^x overflow:

$$\frac{\text{Inf}}{\text{Inf}} = \text{NaN}$$

e^x underflow:

$$\frac{0}{1+0} = 0$$



We need to handle underflow & overflow #

$$\prod_{i=1}^{2000} p(x_i) = \left(\frac{1}{2}\right) \cdots \left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^{2000} = \frac{1}{2^{2000}}$$

$$(10^{-1})^{1000} = 10^{-1000} \quad \text{underflow}$$

$$n! \quad 1000! = 1000 \times 999 \times \cdots \times 1 \quad \text{overflow}$$

$$n! \approx n^{n-\frac{1}{2}}$$

$$\approx 1000^{1000} = 10^{3000} \quad \text{overflow}$$

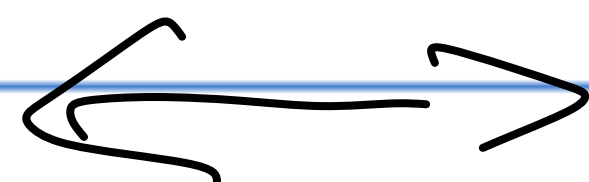
Representing # i's logarithm form

2^{+2000}



$\log_e(x)$
2000

2^{-2000}

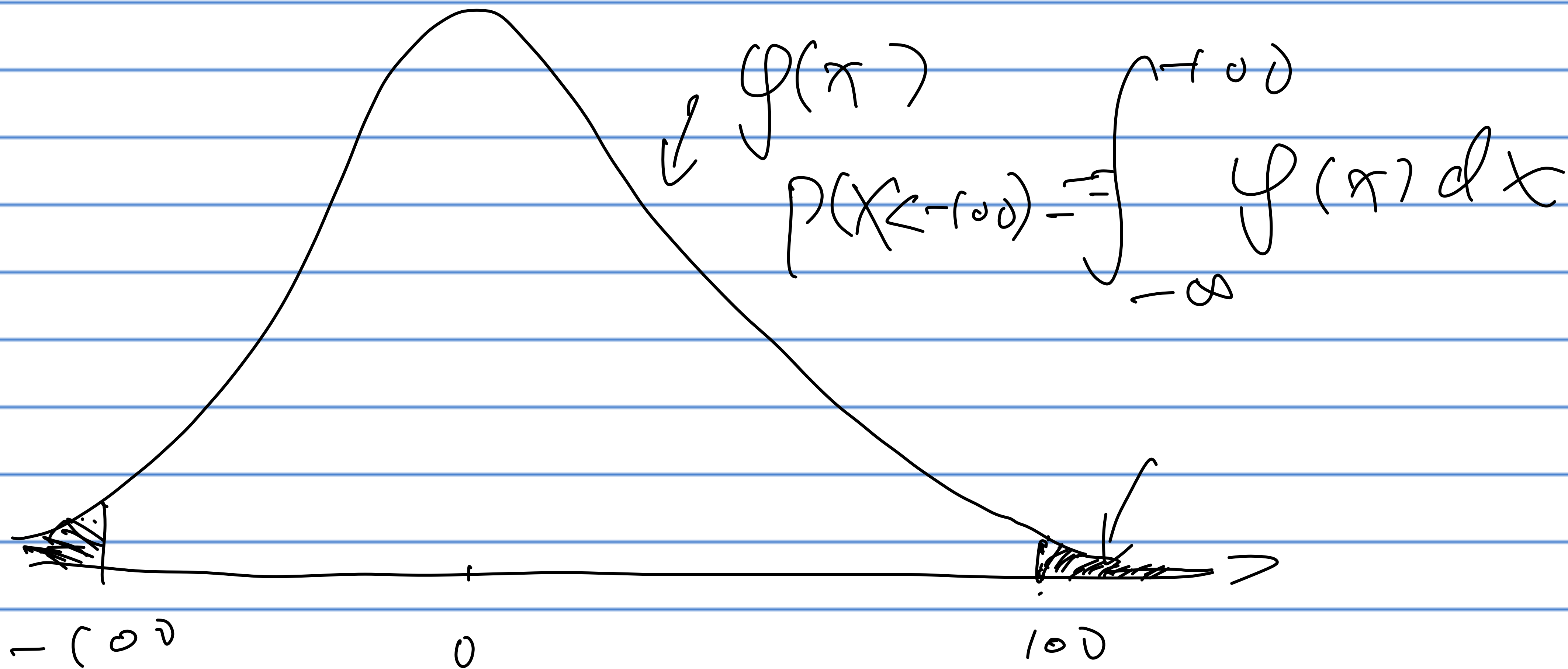


-2000

Given $l_x = \log(x)$, return $\log(f(e^e))$

$x_1, \dots, x_n \xrightarrow{f} y$

$\downarrow \quad \downarrow \quad \downarrow$
 $l_1 \quad l_2 \quad l_y$



Computing $\log(e^{l_1} \cdot e^{l_2})$

$$\log(x_1 \cdot x_2) = \log x_1 + \log x_2, \quad \log x_i = l_i$$

$$= l_1 + l_2$$

$$\log\left(\prod_{i=1}^n x_i\right) = \sum_{i=1}^n \log x_i = \sum_{i=1}^n l_i$$

Example: $\log(n!) = \log(n) + \log(n-1) + \dots + \log(1)$

Computing $\log\left(\frac{x_1}{x_2}\right)$ lfactorial(1000)

$$\log\left(\frac{x_1}{x_2}\right) = \log x_1 - \log x_2 = l_1 - l_2$$

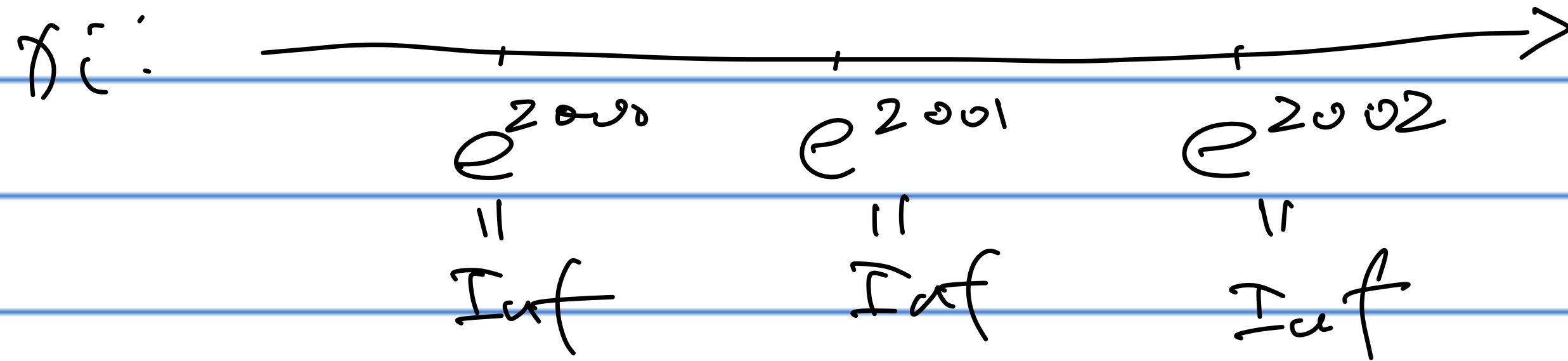
Example: $x_1 = 2^{2000}$, $x_2 = 2^{2001}$, $\log\left(\frac{x_1}{x_2}\right) = \log\left(\frac{1}{2}\right)$

Computing $\log \sum e^{l_i}$:

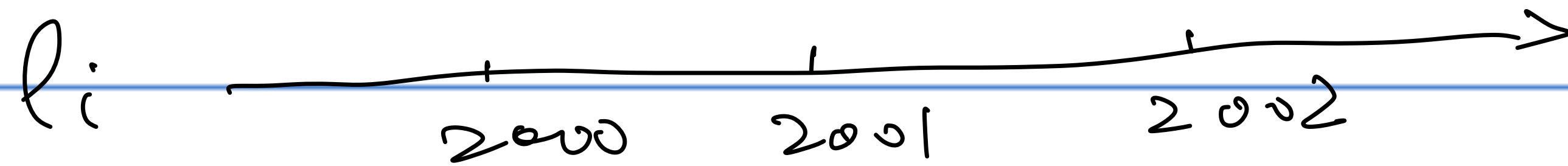
$$\log\left(\sum_{i=1}^n x_i\right) = \log\left(\sum_{i=1}^n e^{l_i}\right) \quad \text{[log-sum-exp]}$$

$x_1 \quad x_2 \quad \dots \quad x_n$
 $\uparrow \quad \uparrow \quad \quad \quad \uparrow$
 $l_1 = \log x_1 \quad l_2 = \log x_2 \quad \quad \quad l_n = \log x_n$

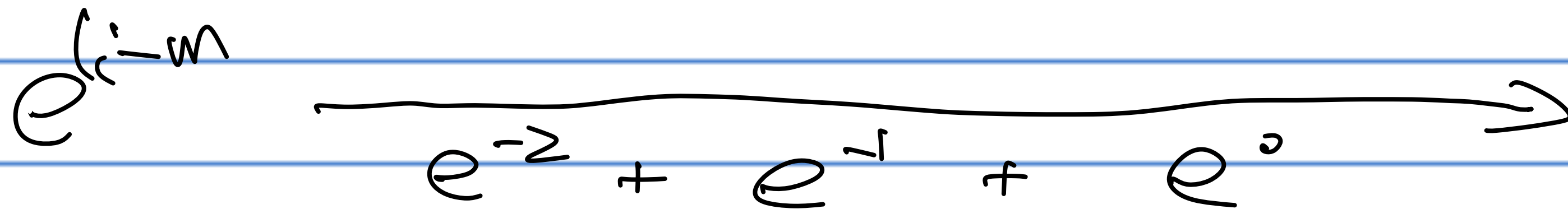
$$\begin{aligned} \log\left(\sum_{i=1}^n x_i\right) &= \log\left(\sum_{i=1}^n e^{l_i}\right) \\ &= \log\left(\sum_{i=1}^n e^{l_i - m} \cdot e^m\right) \\ &= \log\left(\sum_{i=1}^n e^{l_i - m}\right) + m \end{aligned}$$



$$\sum_{i=1}^3 x_i = I_{af}$$



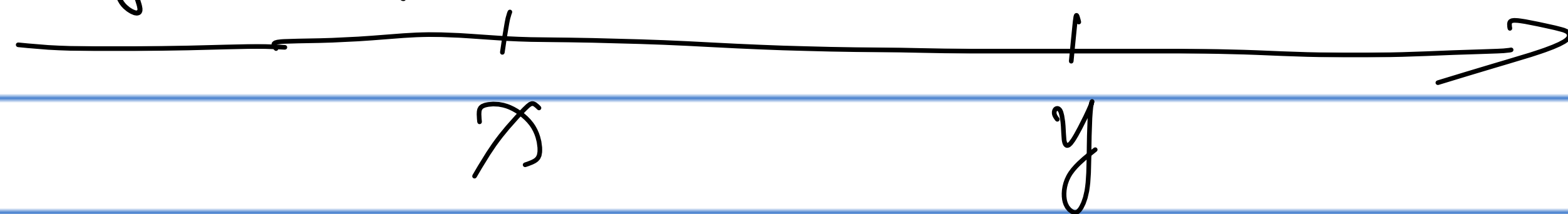
$$m = \max(l_i)$$



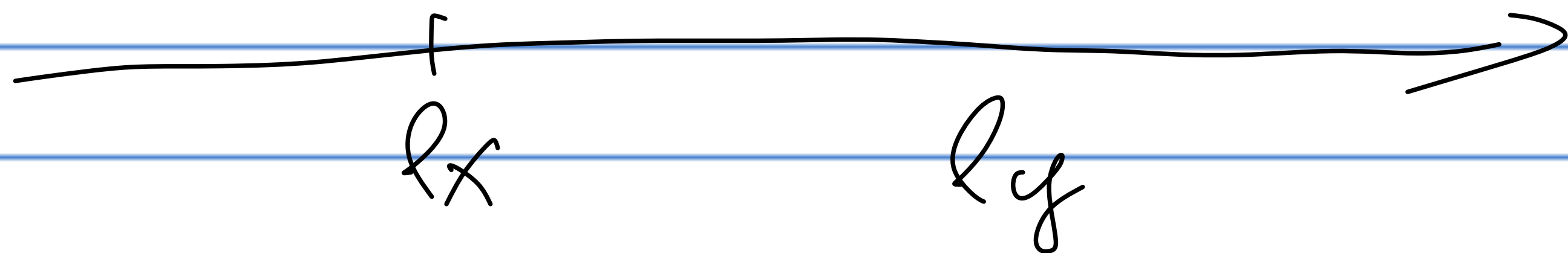
$$\sum_{i=1}^n e^{l_i - m} \rightarrow \log\left(\sum_{i=1}^n e^{l_i - m}\right) \rightarrow \log\left(\sum_{i=1}^n e^{l_i - m}\right) + m$$

$$\log\left(\sum_{i=1}^n e^{x_i}\right)$$

Computing $\log(e^{lx} - e^{ly})$



$$x > y$$



$$\log(x - y) = \log(e^{lx} - e^{ly})$$

$$= \log(e^{lx} (e^{lx-ly} - e^0))$$

$$= lx + \log(1 - e^{ly-lx})$$

Question 1:

How to compute $\log\left(\frac{1}{1+e^{-x}}\right)$

for overflow & underflow e^x .

Question 2: Softmax

$\log\left(\frac{e^{x_k}}{\sum_{k=1}^K e^{x_k}}\right)$, for underflow & overflow e^{x_k}